

ELEC0129 Formula Sheet

Note to students:

1. This is a list of formulae which you do **NOT** need to memorize.
2. However, they will not appear in the exam as a separate page.
3. Instead, they will appear in the particular question where required. For e.g. "Given the rotation matrix, calculate the Euler parameters. Hint: Formula is"

Spacial Description and Transformation:

Equivalent Angle-Axis representation of rotation:

$${}^A_B R = \begin{bmatrix} k_x^2 v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y^2 v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z^2 v\theta + c\theta \end{bmatrix}$$

Where $v\theta = 1 - c\theta$.

Euler Parameters:

$$\varepsilon_1 = k_x \sin(\theta/2), \varepsilon_2 = k_y \sin(\theta/2), \varepsilon_3 = k_z \sin(\theta/2), \varepsilon_4 = \cos(\theta/2)$$

$${}^A_B R = \begin{bmatrix} 1 - 2\varepsilon_2^2 - 2\varepsilon_3^2 & 2(\varepsilon_1\varepsilon_2 - \varepsilon_3\varepsilon_4) & 2(\varepsilon_1\varepsilon_3 + \varepsilon_2\varepsilon_4) \\ 2(\varepsilon_1\varepsilon_2 + \varepsilon_3\varepsilon_4) & 1 - 2\varepsilon_1^2 - 2\varepsilon_3^2 & 2(\varepsilon_2\varepsilon_3 - \varepsilon_1\varepsilon_4) \\ 2(\varepsilon_1\varepsilon_3 - \varepsilon_2\varepsilon_4) & 2(\varepsilon_2\varepsilon_3 + \varepsilon_1\varepsilon_4) & 1 - 2\varepsilon_1^2 - 2\varepsilon_2^2 \end{bmatrix}$$

Forward Kinematics:

Transformation from frame $\{i-1\}$ to $\{i\}$ from the DH-Table:

$${}^{i-1}_i T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverse Kinematics:

None.

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Jacobians:

Velocity propagation:

Start with ${}^0\omega_0, {}^0v_0$.

Calculate recursively from link to link:

- For revolute joints:

$$\begin{aligned} {}^{i+1}\omega_{i+1} &= {}^{i+1}_i R \cdot {}^i\omega_i + \dot{\theta}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1} \\ {}^{i+1}v_{i+1} &= {}^{i+1}_i R \cdot ({}^iv_i + {}^i\omega_i \times {}^iP_{i+1}) \end{aligned}$$

- For prismatic joints:

$$\begin{aligned} {}^{i+1}\omega_{i+1} &= {}^{i+1}_i R \cdot {}^i\omega_i \\ {}^{i+1}v_{i+1} &= {}^{i+1}_i R \cdot ({}^iv_i + {}^i\omega_i \times {}^iP_{i+1}) + \dot{d}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1} \end{aligned}$$

Continue until ${}^n\omega_n, {}^nv_n$.

The velocity of end-effector is thus simply:

$$\begin{aligned} {}^n\omega_e &= {}^n\omega_n \\ {}^nv_e &= {}^nv_n + {}^n\omega_n \times {}^nP_e \end{aligned}$$

Trajectory Planning:

Linear trajectory with parabolic blends:

Calculate $t_b = \frac{t_f}{2} - \frac{\sqrt{\ddot{u}^2 t_f^2 - 4\ddot{u}(u_f - u_0)}}{2\ddot{u}}$ and $u_b = u_0 + \frac{1}{2}\ddot{u}t_b^2$

The trajectory is then:

$$u(t) = \begin{cases} u_0 + \frac{1}{2}\ddot{u}t^2 & t < t_b \\ \frac{u_f - u_b}{t_f - t_b}(t - t_b) + u_b & t_b \leq t < (t_f - t_b) \\ u_f - \frac{1}{2}\ddot{u}(t_f - t)^2 & t \geq (t_f - t_b) \end{cases}$$

Dynamics:

Newton-Euler Formula:

- Outward iteration:

- Start with ${}^0\omega_0 = 0, {}^0\dot{\omega}_0 = 0, {}^0\dot{v}_0 = \text{depends}$

- Calculate velocities and accelerations of frames:

$$\begin{aligned} {}^{i+1}\omega_{i+1} &= ({}^{i+1}_i R \cdot {}^i\omega_i) + (\dot{\theta}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1}) \\ {}^{i+1}\dot{\omega}_{i+1} &= ({}^{i+1}_i R \cdot {}^i\dot{\omega}_i) + (({}^{i+1}_i R \cdot {}^i\omega_i) \times (\dot{\theta}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1})) + (\ddot{\theta}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1}) \\ {}^{i+1}\dot{v}_{i+1} &= ({}^{i+1}_i R \cdot {}^i\dot{v}_i) + (2{}^{i+1}\omega_{i+1} \times (\dot{d}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1})) + (\ddot{d}_{i+1} \cdot {}^{i+1}\hat{Z}_{i+1}) \\ &\quad + {}^{i+1}_i R \left(({}^i\dot{\omega}_i \times {}^iP_{i+1}) + ({}^i\omega_i \times ({}^i\omega_i \times {}^iP_{i+1})) \right) \end{aligned}$$

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- Propagate **accelerations** from frames to **centre of mass**:

$${}^{i+1}\dot{v}_{ci+1} = ({}^{i+1}\dot{v}_{i+1}) + ({}^{i+1}\dot{\omega}_{i+1} \times {}^{i+1}P_{ci+1}) + ({}^{i+1}\omega_{i+1} \times ({}^{i+1}\omega_{i+1} \times {}^{i+1}P_{ci+1}))$$

- Calculate the **force and moment** at the centre of mass:

$$\begin{aligned} {}^{i+1}F_{i+1} &= m_{i+1} {}^{i+1}\dot{v}_{ci+1} \\ {}^{i+1}N_{i+1} &= ({}^{ci+1}I_{i+1} \cdot {}^{i+1}\dot{\omega}_{i+1}) + ({}^{i+1}\omega_{i+1} \times ({}^{ci+1}I_{i+1} \cdot {}^{i+1}\omega_{i+1})) \end{aligned}$$

- Do until the n th link.

- Inward iteration:

- Start with **force and torque at robot tip** ${}^n f_n, {}^n n_n$.

- Calculate force and torque at the starting end of each link:

$$\begin{aligned} {}^i f_i &= ({}_{i+1}^i R \cdot {}^{i+1}f_{i+1}) + {}^i F_i \\ {}^i n_i &= ({}_{i+1}^i R \cdot {}^{i+1}n_{i+1}) + ({}^i P_{ci} \times {}^i F_i) + ({}^i P_{i+1} \times ({}_{i+1}^i R \cdot {}^{i+1}f_{i+1})) + {}^i N_i \end{aligned}$$

- Do until $i = 1$.

- Finally, extract the joint (motor) torques or forces:

$$\begin{aligned} \tau_i &= {}^i n_i^T \cdot {}^i \hat{Z}_i \text{ if revolute} \\ \tau_i &= {}^i f_i^T \cdot {}^i \hat{Z}_i \text{ if prismatic} \end{aligned}$$

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Dynamics in Cartesian Space:

$$\begin{aligned} &J^{-T}(q)M(q)(J^{-1}(q)\ddot{x} - J^{-1}(q)\dot{J}(q)\dot{q}) + J^{-T}(q)V(q, \dot{q}) + J^{-T}(q)G(q) \\ &= \underbrace{J^{-T}M(q)J^{-1}}_{\hat{M}_x} \ddot{x} + \underbrace{J^{-T}(V(q, \dot{q}) - M(q)J^{-1}\dot{J}\dot{q})}_{\hat{V}_x} + \underbrace{J^{-T}G(q)}_{\hat{G}_x} = F \end{aligned}$$

Inertia Tensor:

$${}^A I = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix}$$

- **Mass moment of inertia:**

$$I_{xx} = \iiint_V (y^2 + z^2) \rho dV, I_{yy} = \iiint_V (x^2 + z^2) \rho dV, I_{zz} = \iiint_V (x^2 + y^2) \rho dV$$

- **Mass product of inertia:**

$$I_{xy} = \iiint_V xy \rho dV, I_{xz} = \iiint_V xz \rho dV, I_{yz} = \iiint_V yz \rho dV$$

Control:

None